

Key for Midterm #1

- ① The roots of $x^7 - 2$ are $\sqrt[7]{2} \cdot \alpha^i$ where $\alpha = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ is a primitive 7-th root of 1.

So $\sqrt[7]{2}, \sqrt[7]{2}\alpha \in E$ so $\sqrt[7]{2}, \alpha \in E$ so $E = \mathbb{Q}(\sqrt[7]{2}, \alpha)$

Now $[\mathbb{Q}(\sqrt[7]{2}) : \mathbb{Q}] = 7$ as $x^7 - 2$ is irred by Eisenstein with $p=2$, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 6$ as α is a root of $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$, which is irred as shown in class ($x \rightarrow x+1$ trick + Eisenstein).

But $\text{GCD}(7, 6) = 1$ so $[E : \mathbb{Q}] = 42$.

- ② $[E : \mathbb{Q}] = 2$ then $E = \mathbb{Q}(\alpha)$ and $\text{Irr}(\alpha, \mathbb{Q}, x) = x^2 + bx + c$ which has roots $\frac{-b \pm \sqrt{b^2 - 4ac}}{2}$ let $\Delta = b^2 - 4ac$ then

$\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{\Delta}) = \mathbb{Q}(\pm\sqrt{\Delta})$ so E is normal as it is the splitting field of $x^2 + bx + c$

$\mathbb{Q}(\sqrt[3]{2})$ is not normal as the irred poly of $\sqrt[3]{2}$ is $x^3 - 2$. $\alpha = \sqrt[3]{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$ is another root not in $\mathbb{Q}(\sqrt[3]{2})$.

~~E is algebraic over K so generated by a set $\{\alpha_i\}_{i \in \mathbb{N}}$ of elements~~
~~not algebraic over K . Let $\sigma_1, \dots, \sigma_n$ be the distinct embeddings~~
~~of E in K^n , then $\sigma_1 E, \sigma_2 E, \dots, \sigma_n E$ is the smallest normal subfield~~
~~of K^n containing E .~~

- ③ Let $y^p - x^p \in \mathbb{F}_p[x^p]$, then $y^p - x^p = (y - x)^p$ has a repeated root (namely x) with multiplicity p . Let $f(y) = \text{irr}(x, \mathbb{F}_p[x^p], y)$, then $f(y)$ divides $(y - x)^p$ and so $f(y)$ has a unique (repeated) root. It follows that $[\mathbb{F}_p(x) : \mathbb{F}_p(x^p)] = 1$.

- F) Let $\phi: K[x] \rightarrow K[\alpha] \subset K$ be the natural homomorphism such that $\phi(x) = \alpha$. $K[x]$ is a PID so $\ker \phi = (p(x))$ &

$K[\alpha] = K[x]/(p(x)) \subset K$ is a domain so $p(x)$ is irreducible.

Let $af(\alpha) \in K[\alpha]$, we must show that $g(\alpha)$ is invertible in $K[\alpha]$.

Since $0 \neq g(x)$, one has that $g(x) \nmid (p(x))$ and since $p(x)$ is irreducible, then $\text{GCD}(g(x), p(x)) = 1$. So we

may write $1 = g(x)h(x) + p(x)k(x)$. Then

$$1 = g(x)h(x) + p(x)k(x) = g(x)h(x) \quad (\text{as } p(x) = 0).$$

So $g(x)$ is invertible.